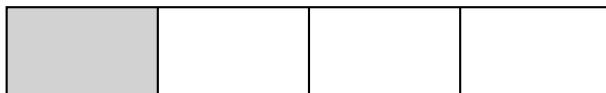


Fraction Models: Are the Parts Equal?

Remember: a fraction only names the parts of a whole when *every part is the same size*. Cutting a shape into four pieces does not make one piece a fourth — the four pieces must all be the same size first.

Each picture below is somebody's model. Some of them are wrong. Show your thinking in the work space, then write your final answer on the answer line.

1. Look at this model. All of its parts are the same size.



Are all of the parts the same size? Write the fraction of the whole rectangle that is shaded.

Answer: _____

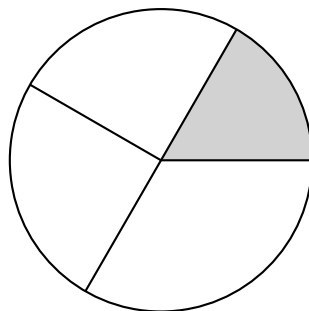
2. Ben broke this strip along the heavy lines. The thin lines show that the strip is made of 6 equal small parts.



Ben says the shaded piece is $\frac{1}{3}$ of the strip, because it is 1 of his 3 pieces. Explain Ben's mistake in the work space. Then write the fraction of the strip that is really shaded.

Answer: _____

3. Kim cut this pizza into 4 pieces and shaded one of them.



Kim says the shaded piece is $\frac{1}{4}$ of the pizza. **(a)** Write one sentence saying why Kim is wrong. **(b)** In the space below, draw the same pizza cut into fourths and shade $\frac{1}{4}$ of it.

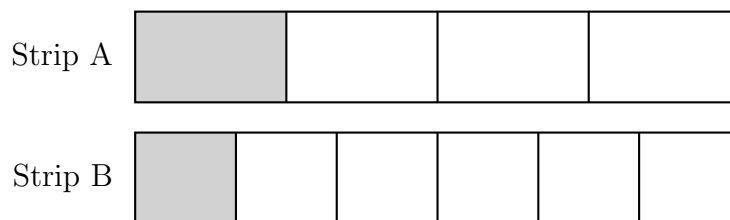
4. Nina broke this strip along the heavy lines. The thin lines show that the strip is made of 8 equal small parts.



Nina says the shaded piece is $\frac{1}{3}$ of the strip, because it is 1 of her 3 pieces. Explain Nina's mistake in the work space. Then write the fraction of the strip that is really shaded.

Answer: _____

5. These two paper strips are exactly the same length. Strip A is folded into equal parts, and so is Strip B. One part of each strip is shaded.



The shaded part of Strip A is $\frac{1}{4}$ and the shaded part of Strip B is $\frac{1}{6}$. Which shaded part is larger? Write $>$ or $<$ to make the sentence true: $\frac{1}{4}$ _____ $\frac{1}{6}$.

Answer: _____

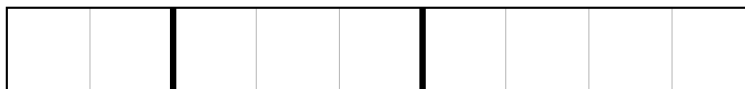
6. Tara wants to fold this strip of 8 equal squares into 2 equal halves. She drew her fold line after 3 squares and shaded that piece.



Explain in the work space why Tara's two pieces are not halves. Then write how many squares should be in each half.

Answer: _____

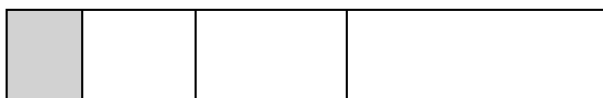
- 7.** Three friends want to share this cracker bar fairly. The bar is made of 9 equal squares. Cal broke it into the 3 pieces shown.



Explain in the work space why Cal's pieces are not a fair share. Then write how many squares each friend should get so that every share is $\frac{1}{3}$ of the bar.

Answer: _____

- 8.** Zoe says: “*Whenever a shape is cut into 4 pieces, one piece is $\frac{1}{4}$ of the shape.*” Here is one of Zoe's models.



- (a)** Use one sentence to explain why Zoe is wrong. **(b)** Below, draw your own model of the same rectangle that really does show $\frac{1}{4}$, and shade one fourth.